EuroSAT Land-Cover Classifier

An original CNN pipeline for recognizing landscapes in Sentinel-2 satellite imagery

10 classes Agriculture, forest, vegetation, roads, industry, residential areas, rivers and water bodies	**87.6%** measured test accuracy **87.4%** measured macro-F1	**27,000 patches** Public RGB image archive **3 bands** RGB input

Abstract

This project classifies a 64 × 64 RGB satellite patch into one of ten land-use/land-cover categories. A compact convolutional neural network (CNN), written specifically for this project and trained from random initialization, learns visual patterns in the EuroSAT dataset. The pipeline downloads and checksum-verifies the public archive, creates stratified train/validation/test splits, trains and selects a checkpoint using validation macro-F1, reports class-wise performance on held-out test images, and produces confusion-matrix and Grad-CAM visualizations. The results below come from the local run described on page 2; they are not copied from the benchmark paper or any GitHub repository.

Dataset and source

EuroSAT is a Sentinel-2 land-use and land-cover dataset with 27,000 labeled, georeferenced image patches across 10 categories. The original dataset contains 13 spectral bands; this implementation uses the readily available RGB archive (three channels, about 94.7 MB). The downloaded archive was checked against Zenodo's published MD5 checksum before extraction. The original paper reports the benchmark and describes both RGB and multispectral data. Dataset download: Zenodo record 7711810 (DOI 10.5281/zenodo.7711810). The dataset authors mark it MIT-licensed; Sentinel terms also apply.

Problem framing

Satellite scenes have no canonical orientation, so the training pipeline uses horizontal and vertical flips. The target is the patch-level land-cover label, not pixel segmentation or a complete geographic map. The random image split supports a benchmark-style evaluation but does not prove generalization to unseen regions.



Example EuroSAT RGB patches: forest, residential, and river.

Method and experiment

A small, reproducible pipeline designed to run on CPU or GPU

RGB patch
64 × 64 × 3
→
Augmentation
horizontal / vertical flips
→
CNN
three convolutional blocks
→
Class probabilities
10 land-cover categories

Model details

The network begins with a stride-2 3 × 3 convolution and batch normalization, followed by three two-convolution blocks with 32, 64, and 128 channels. Each convolution is followed by batch normalization and ReLU; each block ends in 2 × 2 max-pooling. Adaptive average pooling reduces the final feature map to one spatial cell. A dropout layer and linear 10-class head produce logits. Total trainable parameters: **295,490**. Optimization uses AdamW, cross-entropy loss, learning rate 0.001, and weight decay 0.0001. No pretrained model or external checkpoint is used.

Local run configuration

A fixed seed (42) generated class-stratified partitions from all 27,000 images: 18,900 in the training pool, 4,050 validation, and 4,050 held-out test. For this local run, 10,000 of the 18,900 training-pool images (1000 per class) were used. Training ran for 8 epochs with batches of 64 on cpu (PyTorch 2.15.0+cpu). The best checkpoint was selected by validation macro-F1 at epoch 7.

Reproduce from the project folder

1. Install `requirements.txt` in a Python 3.10+ environment. 2. Run `python download_data.py` to download, checksum-check, and extract EuroSAT RGB. 3. Run `python train.py --data data/EuroSAT_RGB --epochs 8 --batch-size 64 --train-per-class 1000 --threads 4 --cpu`. 4. Run `python predict.py --checkpoint outputs/best_model.pt` for a prediction and Grad-CAM. The dataset archive is public and about 95 MB; model training is slower on CPU. Omit `--train-per-class` to use the full training pool.

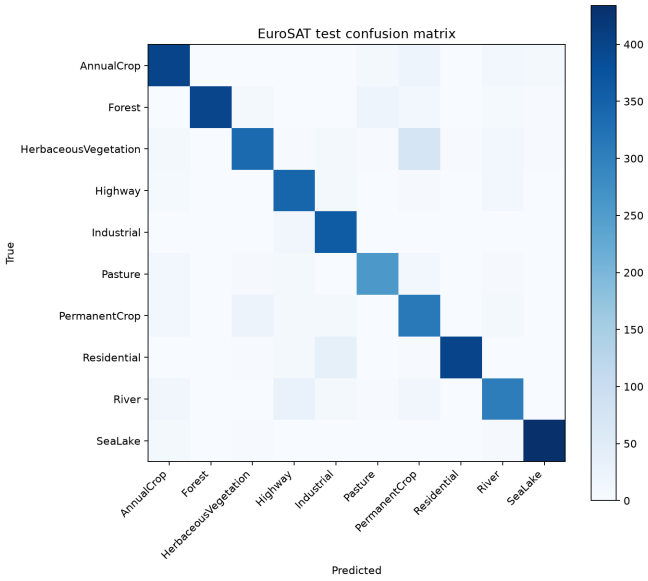
Deliverables from the run include the checkpoint, split manifest, per-class JSON metrics, training history, confusion matrix, and learning curve. The PDF report is generated from `outputs/metrics.json`, preventing published benchmark scores from being substituted for the local results.

Measured results

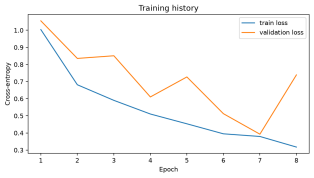
Held-out test split from the local CPU training run

TEST ACCURACY	MACRO-F1	TEST IMAGES	BEST EPOCH
87.6%	**87.4%**	**4,050**	**7**

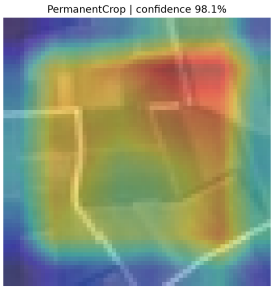
Class	Prec.	Rec.	F1	N
AnnualCrop	0.87	0.89	0.88	450
Forest	0.99	0.88	0.94	450
HerbaceousVegetation	0.88	0.75	0.81	450
Highway	0.82	0.92	0.87	375
Industrial	0.84	0.96	0.89	375
Pasture	0.88	0.86	0.87	300
PermanentCrop	0.69	0.83	0.75	375
Residential	0.99	0.89	0.94	450
River	0.84	0.81	0.83	375
SeaLake	0.97	0.96	0.97	450



Reading the result. The dataset is publicly available and the test set was not used for training or model selection. This run used 10,000 of the 18,900 training-pool images for 8 epochs. The measured score is therefore an early local baseline, not a reproduction of the paper's 98.57% benchmark. More training data, longer schedules, geographic splits, and additional seeds are needed for a stronger generalization claim.



Training loss



Grad-CAM

Base paper and attribution

Helber, P., Bischke, B., Dengel, A., & Borth, D. (2019). “EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification.” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7), 2217–2226. https://doi.org/10.1109/JSTARS.2019.2918242. Preprint: https://arxiv.org/abs/1709.00029. Dataset DOI: https://doi.org/10.5281/zenodo.7711810. This project uses RGB data and an independently implemented CNN; the paper's benchmark is reported only as context, not as our result.